BEN ESTRADA

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SUMMARY

Set Lighting Technician with 20+ years of experience in precision imaging, lighting, and electrical systems for high-end film and television, now applying quantum and photonics training to quantum-optical experiments and R&D lab work. Completed CNM Ingenuity's full-time Quantum Technician Boot Camp with intensive hands-on work in lasers, optics/photonics, ultrahigh vacuum systems, spectroscopy, interferometry, and quantum optics experiments. Strong track record building and aligning complex technical setups from drawings and procedures, with meticulous organization, clear documentation, and collaborative troubleshooting in safety-critical, fast-paced environments.

SKILLS

Quantum, Optics, and Photonics: Free-space optical alignment and beam steering, Laser alignment and tuning; Class 3B/4 laser safety, Optical assemblies using lenses, mirrors, waveplates, beam splitters, diffraction gratings, polarizing optics, Optical fiber coupling and routing; basic fiber-optic and electrical cable building, Use of spectrometers, beam profilers, and photodetectors

Vacuum and Experimental Systems: UHV component assembly and bake-out; contamination control practices, Helium leak detection support; pump operation and monitoring (turbo, ion, wet, scroll), Vacuum troubleshooting and basic diagnostics

Electronics, Build, and Tools: Soldering/PCB soldering (through-hole and SMT), PCB assembly and rework, Oscilloscopes, digital multimeters, basic power supplies, SolidWorks, 3D-printed fixtures, simple mechanical fabrication

Lab and Team Practices: SOPs, test checklists, build logs, lab notebooks, Lean/5S work practices and workspace organization, Clear written and verbal communication; effective work in diverse, interdisciplinary teams

EDUCATION

Quantum Technician | CNM Ingenuity | Albuquerque, New Mexico | 2026

- Lab-intensive program in optics and photonics, lasers and detectors, ultrahigh vacuum systems, and quantum phenomena/quantum information systems.
- Built and aligned experiments including saturated absorption spectroscopy, Michelson interferometry, optical tweezers, quantum key distribution-style setups, and single-photon interferometry.
- Assembled mechanical, optical, and basic electrical components using posts, mounts, lenses, mirrors, wave plates, beam splitters, diffraction gratings, and fiber couplers.
- Worked with UHV assembly and bake-out, helium leak detection, and operation of turbo, ion, wet, and scroll pumps; followed detailed procedures and maintained clear lab notebooks.
- Used oscilloscopes, digital multimeters, and basic test equipment to support optical experiments and control electronics.

Bachelor of Arts, Technical Theatre | University of New Mexico | Minor in Film and Television Production | Albuquerque, New Mexico | 1997

- Minor in Film and Television Production.

CERTIFICATIONS

Quantum Technician Certificate | CNM Ingenuity | 2026

Lean Manufacturing Fundamentals | New Mexico Manufacturing Extension Partnership | 2026

IPC/EIA J-STD-001 Soldering Certification | Electronic Workmanship Standards Institute | 2026

EXPERIENCE

Technical Assembly and Handyman (Task-Based Work) | Various Clients | Albuquerque, New Mexico | May 2026 - Present

- Assembled complex boxed furniture and technical fixtures from drawings and instructions, ensuring clean, repeatable builds and resolving issues independently.
- Installed and debugged basic home electrical and plumbing components using safe work practices and test equipment.
- Demonstrated reliability in following work instructions and diagrams, building safe assemblies, and communicating with clients, similar to building, wiring, and maintaining complex experimental setups in a lab.

Set Lighting Technician / Assistant Chief Lighting Technician (Best Boy) | IATSE Studio Mechanics Local 480 | Albuquerque / Santa Fe, New Mexico | January 2003 - May 2025

- Executed precision lighting, imaging, and power-distribution setups for large-scale film and television productions in fast-paced, technically demanding environments.
- Installed, aligned, and operated tungsten, LED, HMI, and specialty lighting systems requiring precise beam control, focus, and uniformity, directly analogous to aligning lasers and free-space optics.
- Interpreted technical drawings, lighting plots, and power diagrams to design and build complex electrical and rigging systems; managed electrical load balancing, troubleshooting, and equipment diagnostics under tight time constraints.
- Performed safety inspections and maintained compliance with electrical, rigging, elevated-work, and set safety standards, routinely handling heavy equipment and operating aerial lifts.
- Organized and maintained large inventories of fixtures, optics, cable, and hardware using 5S-style methods to keep workspaces clean, safe, and efficient.
- Collaborated with multidisciplinary teams to plan, build, and maintain technical setups across multiple locations per day on productions such as Pluribus, Better Call Saul, The Avengers, Breaking Bad, and No Country for Old Men.